

DAY 23

A/B Testing and Hypothesis Testing in Practice

Prepared for Muhammad Ahmad

60-Day Data Science Learning Journey • Python • SciPy • NumPy • Data-driven decisions

Learning outcomes

- Explain control and treatment groups and conversion rates.
- Form null and alternative hypotheses.
- Interpret significance levels and p-values.
- Run a chi-square test with SciPy.
- Validate data and communicate results responsibly.

1. A/B Testing Fundamentals

A/B testing compares two versions using a predefined metric. Version A is commonly the control, while Version B is the alternative or treatment. For a website checkout, the metric could be the proportion of visitors who purchase.

Example: A receives 1,000 visitors and 120 purchases. B receives 1,000 visitors and 150 purchases. A's conversion rate is 12%; B's is 15%. The observed difference is 3 percentage points. This difference alone does not prove the population conversion rates differ.

Conversion rate = purchases / visitors × 100. A good experiment uses a clear metric, random assignment where feasible, comparable groups, a planned analysis, and enough observations to detect a meaningful effect.

2. Key Statistical Concepts

Concept	Meaning
Null hypothesis (H_0)	Population conversion rates are equal.
Alternative hypothesis (H_1)	Population conversion rates differ in a two-sided test.
Significance level (α)	Threshold chosen before testing, often 0.05.
P-value	Assuming H_0 and test assumptions hold, probability of a result at least as extreme as observed.
Type I error	Rejecting a true null hypothesis.
Type II error	Failing to reject a false null hypothesis.

Important: A p-value is not the probability that H_1 is true. Statistical significance does not automatically mean the effect is large, useful, or causal.

3. Calculate Conversion Rates

```
visitors_a = 1000
purchases_a = 120
visitors_b = 1000
purchases_b = 150

rate_a = purchases_a / visitors_a
rate_b = purchases_b / visitors_b

print(f"Version A: {rate_a:.1%}")
print(f"Version B: {rate_b:.1%}")
print(f"Difference: {rate_b - rate_a:.1%}")
```

Line by line:

- The visitor variables store the number of visitors in each group.
- The purchases variables store successful conversions.
- Dividing purchases by visitors gives a proportion between 0 and 1.
- The .1% format displays a percentage with one decimal place.
- Subtracting 12% from 15% gives 3 percentage points, not a 3% relative increase.

4. Run a Chi-Square Test with SciPy

For two independent groups and a binary outcome (purchase or no purchase), a chi-square test of independence is a common choice when expected cell counts are sufficiently large. A two-proportion z-test is another option. For a 2×2 table, Pearson's chi-square statistic without continuity correction equals the pooled two-proportion z statistic squared.

```
import numpy as np
from scipy.stats import chi2_contingency

visitors_a, purchases_a = 1000, 120
visitors_b, purchases_b = 1000, 150

rate_a = purchases_a / visitors_a
rate_b = purchases_b / visitors_b

table = np.array([
    [purchases_a, visitors_a - purchases_a],
    [purchases_b, visitors_b - purchases_b]
])

chi2, p_value, dof, expected = chi2_contingency(
    table, correction=False
)

print(f"Version A conversion rate: {rate_a:.1%}")
print(f"Version B conversion rate: {rate_b:.1%}")
print(f"Observed difference: {rate_b - rate_a:.1%}")
print(f"Chi-square statistic: {chi2:.4f}")
print(f"P-value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis.")
    print("There is evidence of a difference.")
else:
    print("Fail to reject the null hypothesis.")
    print("The evidence is insufficient.")
```

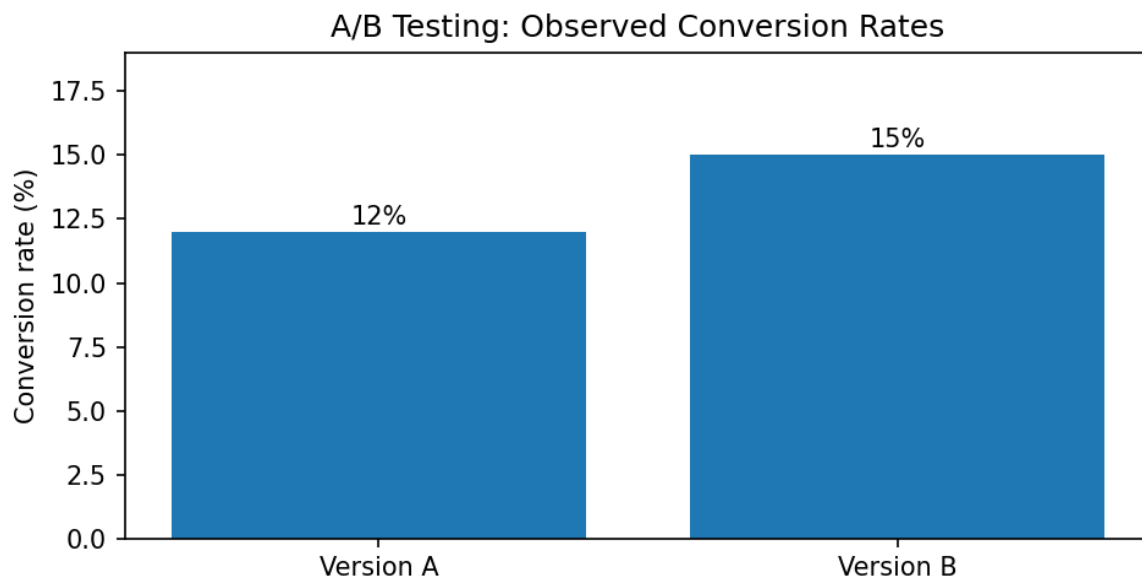
Expected output:

```
Version A conversion rate: 12.0%
Version B conversion rate: 15.0%
Observed difference: 3.0%
Chi-square statistic: 4.9206
P-value: 0.0265
Reject the null hypothesis.
There is evidence of a difference.
```

Code explanation:

- NumPy constructs a 2×2 table. Each row is a version; columns are purchases and non-purchases.
- `chi2_contingency` compares observed counts with expected counts under the null of independence.
- The function returns the chi-square statistic, p-value, degrees of freedom, and expected counts.
- Since $p \approx 0.0265$ is below $\alpha = 0.05$, reject H_0 for this two-sided test.
- This is evidence of a difference, not proof that B will always win. Check experiment design and practical value.

5. Visualize the Results



The chart describes the observed rates. It is not itself a significance test. Confidence intervals can help communicate uncertainty.

6. Complete A/B Testing Analyzer Project

This project validates input counts, computes rates, runs the test, prints a cautious conclusion, and plots the results.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2_contingency

def get_valid_counts(group_name):
    visitors = int(input(f"{group_name} visitors: "))
    purchases = int(input(f"{group_name} purchases: "))

    if visitors <= 0:
        raise ValueError("Visitors must be greater than zero.")
    if purchases < 0 or purchases > visitors:
        raise ValueError("Purchases must be between 0 and visitors.")

    return visitors, purchases

visitors_a, purchases_a = get_valid_counts("Version A")
visitors_b, purchases_b = get_valid_counts("Version B")

rate_a = purchases_a / visitors_a
rate_b = purchases_b / visitors_b
difference = rate_b - rate_a

table = np.array([
    [purchases_a, visitors_a - purchases_a],
    [purchases_b, visitors_b - purchases_b]
])
chi2, p_value, dof, expected = chi2_contingency(
    table, correction=False
)
alpha = 0.05

print(f"Version A: {rate_a:.2%}")
print(f"Version B: {rate_b:.2%}")
print(f"Difference (B - A): {difference:.2%}")
print(f"Chi-square: {chi2:.4f}")
print(f"P-value: {p_value:.4f}")

if p_value < alpha:
    print("Statistically significant difference at alpha = 0.05.")
else:
    print("No statistically significant difference at alpha = 0.05.")

if difference > 0:
    print("Version B has the higher observed conversion rate.")
elif difference < 0:
```

```

    print("Version A has the higher observed conversion rate.")
else:
    print("Both versions have equal observed conversion rates.")

plt.bar(["Version A", "Version B"], [rate_a * 100, rate_b * 100])
plt.ylabel("Conversion rate (%)")
plt.title("A/B Testing Analyzer")
plt.tight_layout()
plt.show()

```

Project line-by-line guide:

- `get_valid_counts()` asks for visitors and purchases and returns validated counts.
- visitors must be positive; purchases must be between zero and visitors for this unique-visitor conversion metric.
- The table counts successes and failures for both groups.
- The test evaluates evidence of a difference; alpha should be selected before examining results.
- The comparison reports which version has the higher observed rate, which is not a guarantee about future performance.
- For production use, plan sample size, avoid repeated unplanned peeking, and review confidence intervals, costs, and guardrail metrics.

7. Sample Project Run

```

Version A visitors: 1000
Version A purchases: 120
Version B visitors: 1000
Version B purchases: 150

```

```

Version A: 12.00%
Version B: 15.00%
Difference (B - A): 3.00%
Chi-square: 4.9206
P-value: 0.0265
Statistically significant difference at alpha = 0.05.
Version B has the higher observed conversion rate.

```

Business interpretation: B's observed conversion rate is 15% versus 12% for A, an increase of 3 percentage points. The test finds evidence of a difference at the 5% level. Before rollout, verify randomization, sample planning, guardrail metrics, implementation costs, and practical value.

8. Practice Questions with Complete Solutions

Task 1: Calculate rates

A gets 500 visitors and 50 purchases. B gets 500 visitors and 65 purchases.

Solution: $A = 50/500 = 10\%$; $B = 65/500 = 13\%$; difference = 3 percentage points.

Task 2: Interpret $p = 0.18$

Solution: $0.18 > 0.05$, so fail to reject H_0 . Evidence is insufficient at the 5% threshold. This does not prove equality.

Task 3: Interpret $p = 0.01$

Solution: $0.01 < 0.05$, so reject H_0 . There is statistically significant evidence of a difference under the test assumptions.

Task 4: Statistical versus practical significance

Solution: A small effect can be statistically significant with a large sample but still fail to justify development or maintenance costs.

Task 5: Why randomize?

Solution: Random assignment helps make groups comparable and reduces selection bias.

Task 6: Validate input

Solution: In a unique-visitor conversion setup, purchases counted as converting visitors cannot exceed visitor count. If repeat purchases are measured, define a different metric and model.

Task 7: Choose the metric before the test

Solution: A predefined primary metric reduces cherry-picking a metric that looks significant after inspecting results.

Task 8: Avoid repeated peeking

Solution: Repeatedly checking a standard p-value and stopping at $p < 0.05$ can increase false-positive risk. Prespecify stopping rules or use sequential methods.

Task 9: Interpret $p = 0.08$

Solution: The result does not cross the 0.05 threshold. It does not prove the versions are identical; examine power and confidence intervals.

Task 10: Write a business conclusion

Solution: "Version B converted 15% of visitors versus 12% for A, an observed increase of 3 percentage points. The two-sided test yielded $p \approx 0.0265$ at $\alpha = 0.05$. This is evidence of a difference, but experiment quality, costs, and guardrail metrics should be checked before rollout."

9. Assumptions and Responsible Interpretation

- Groups should be independent, and the outcome must be clearly defined.
- Random assignment is strongly preferred for causal interpretation.
- Expected counts should be adequate for the chi-square approximation; with sparse counts, consider an exact test or suitable alternative.
- A two-sided test checks for a difference in either direction. Choose a one-sided test only in advance when justified.
- Report effect size and uncertainty, not only a p-value.
- From 12% to 15% is +3 percentage points and a 25% relative increase.

10. Quick Revision

- A/B testing compares variants using a defined outcome.
- H_0 represents no population difference in conversion rates.
- The p-value is conditional on H_0 and the test assumptions.
- If $p < \alpha$, reject H_0 ; otherwise, fail to reject H_0 .
- Statistical significance is not certainty or business importance.
- Good experiments need data quality, randomization, planned sample size, and transparent reporting.

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